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Inventor(s)

Rogers; James William et al.

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### **Aerosol delivery device including connector comprising extension and receptacle**

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#### **Abstract**

The present disclosure relates to an aerosol delivery device. The aerosol delivery device may include a control body with a first connector portion and a cartridge with a second connector portion. The first connector portion and the second connector portion may be configured to releasably engage each other. One of the first connector portion and the second connector portion may include an extension and the other of the first connector portion and the second connector portion may include a receptacle configured to receive the extension. The extension may include contact sections positioned along a longitudinal length thereof. The contact sections may be electrically insulated from one another by at least one spacer and may be configured to form an electrical connection with the receptacle. A related assembly method is also provided.

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**Inventors:** Rogers; James William (Cornelius, NC), Worm; Steven L. (Raleigh, NC), Christopherson; David G. (Raleigh, NC)

**Applicant:** RAI Strategic Holdings, Inc. (Winston-Salem, NC)

**Family ID:** 58448589

**Assignee:** RAI Strategic Holdings, Inc. (Winston-Salem, NC)

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*Primary Examiner:* Paik; Sang Y

*Attorney, Agent or Firm:* Womble Bond Dickinson (US) LLP

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/422,026, filed May 24, 2019, which is a divisional of U.S. application Ser. No. 15/081,485, filed Mar. 25, 2016, which applications are hereby incorporated by reference in their entirety in this application.

### **BACKGROUND**

#### **Field of the Disclosure**

(1) The present disclosure relates to aerosol delivery devices such as electronic cigarettes, and more particularly to connectors for aerosol delivery devices including an atomizer, and associated systems and apparatuses. The atomizer may be configured to heat an aerosol precursor composition, which may be made or derived from tobacco or otherwise incorporate tobacco, to form an inhalable substance for human consumption.

#### **Description of Related Art**

(2) Many smoking devices have been proposed through the years as improvements upon, or alternatives to, smoking products that require combusting tobacco for use. Many of those devices purportedly have been designed to provide the sensations associated with cigarette, cigar, or pipe smoking, but without delivering considerable quantities of incomplete combustion and pyrolysis products that result from the burning of tobacco. To this end, there have been proposed numerous smoking products, flavor generators, and medicinal inhalers that utilize electrical energy to vaporize or heat a volatile material, or attempt to provide the sensations of cigarette, cigar, or pipe smoking without burning tobacco to a significant degree. See, for example, the various alternative smoking articles, aerosol delivery devices and heat generating sources set forth in the background art described in U.S. Pat. No. 8,881,737 to Collett et al., U.S. Pat. App. Pub. No. 2013/0255702 to Griffith Jr. et al., U.S. Pat. App. Pub. No. 2014/0000638 to Sebastian et al., U.S. Pat. App. Pub. No. 2014/0096781 to Sears et al., U.S. Pat. App. Pub. No. 2014/0096782 to Ampolini et al., and U.S. Pat. App. Pub. No. 2015/0059780 to Davis et al., which are incorporated herein by reference in their entireties. See also, for example, the various embodiments of products and heating configurations described in the background sections of U.S. Pat. No. 5,388,594 to Counts et al. and U.S. Pat. No. 8,079,371 to Robinson et al., which are incorporated by reference in their entireties.

(3) However, some aerosol delivery device may include multiple pieces, which may be separable. For example, aerosol delivery devices may include a control body and a cartridge. Accordingly, aerosol delivery devices may include couplers that allow for coupling and decoupling of the cartridge and the control body, such that the cartridge may be refilled or replaced. However, such connectors may wear out from repeated usage, may be difficult to engage or disengage, or may

provide unreliable electrical connections. Thus, advances with respect to connectors for aerosol delivery devices may be desirable.

#### BRIEF SUMMARY OF THE DISCLOSURE

(4) The present disclosure relates to assembly of cartridges for aerosol delivery devices configured to produce aerosol and which aerosol delivery devices, in some embodiments, may be referred to as electronic cigarettes. As described in detail herein, embodiments of connectors for aerosol delivery devices are disclosed. For example, in one aspect, an aerosol delivery device is provided. The aerosol delivery device may include a control body including an electrical power source and a first connector portion. The aerosol delivery device may additionally include a cartridge. The cartridge may include a reservoir configured to contain an aerosol precursor composition, an atomizer configured to heat the aerosol precursor composition received from the reservoir to produce an aerosol, and a second connector portion. The first connector portion and the second connector portion may be configured to releasably engage each other. One of the first connector portion and the second connector portion may include an extension and the other of the first connector portion and the second connector portion may include a receptacle configured to receive the extension. The extension may include a plurality of contact sections positioned along a longitudinal length thereof. The contact sections may be electrically insulated from one another by at least one spacer and may be configured to form an electrical connection with the receptacle.

(5) In some embodiments the contact sections may include a data contact section configured to form a data connection between the cartridge and the control body. The air inlet may be defined in the cartridge. The first connector portion may define a pressure port configured to be in fluid communication with the air inlet when the first connector portion engages the second connector portion.

(6) In some embodiments the first connector portion may further include an O-ring configured to engage an inner surface of the second connector portion. The extension may include a detent and the receptacle may include a flexible member configured to engage the detent to retain the connection between the first connector portion and the second connector portion. The extension may include a tip-ring-sleeve plug. The receptacle may include a center pin terminal. The extension and the receptacle may be centrally disposed with respect to a respective one of the first connector portion and the second connector portion. The extension may be configured to engage the receptacle regardless of a relative rotational position of the cartridge with respect to the control body.

(7) In an additional aspect, a method for assembling an aerosol delivery device is provided. The method may include forming a control body by inserting an electrical power source into a control body outer body and engaging a first connector portion with the control body outer body. Additionally, the method may include forming a cartridge by inserting a reservoir and an atomizer into a cartridge outer body and engaging a second connector portion with the cartridge outer body. The reservoir may be configured to contain an aerosol precursor composition and the atomizer may be configured to heat the aerosol precursor composition received from the reservoir to produce an aerosol. The first connector portion and the second connector portion may be configured to releasably engage each other. One of the first connector portion and the second connector portion may include an extension and the other of the first connector portion and the second connector portion may include a receptacle configured to receive the extension. The extension may include a plurality of contact sections positioned along a longitudinal length thereof. The contact sections may be electrically insulated from one another by at least one spacer and may be configured to form an electrical connection with the receptacle.

(8) In some embodiments engaging the first connector portion with the control body outer body may include engaging a coupler with a flow tube and engaging the flow tube with the control body outer body. Engaging the second connector portion with the cartridge outer body may include engaging a base with the cartridge outer body. Forming the control body may further include



engaging an O-ring with the coupler. The O-ring may be configured to engage an inner surface of the second connector portion.

(9) In some embodiments forming the control body may further include inserting a flow sensor in the control body outer body. The coupler may define a pressure port configured to be in fluid communication with the cartridge when the first connector portion engages the second connector portion. The method may additionally include engaging a pressure tube with the flow sensor and with the coupler.

(10) In some embodiments the method may further include engaging the extension with one of the coupler and the base and engaging the receptacle with the other of the coupler and the base. Engaging the extension with one of the coupler and the base and engaging the receptacle with the other of the coupler and the base may include centrally disposing the extension and the receptacle with respect to a respective one of the coupler and the base. Engaging the extension with one of the coupler and the base may include engaging a tip-ring-sleeve plug with one of the coupler and the base. Additionally, the method may include inserting a controller into the control body outer body and inserting an electronic control component into the cartridge outer body. The method may further include electrically coupling a data contact section of the extension with one of the controller and the electronic control component.

(11) These and other features, aspects, and advantages of the disclosure will be apparent from a reading of the following detailed description together with the accompanying drawings, which are briefly described below.

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## Description

### BRIEF DESCRIPTION OF THE FIGURES

(1) Having thus described the disclosure in the foregoing general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. 1 illustrates a side view of an aerosol delivery device comprising a cartridge and a control body in an assembled configuration according to an example embodiment of the present disclosure;

(3) FIG. 2 illustrates the control body of FIG. 1 in an exploded configuration according to an example embodiment of the present disclosure;

(4) FIG. 3 illustrates the cartridge of FIG. 1 in an exploded configuration according to an example embodiment of the present disclosure;

(5) FIG. 4 illustrates an end view of a base of the cartridge of FIG. 1 according to an example embodiment of the present disclosure;

(6) FIG. 5 illustrates an end view of a coupler of the control body of FIG. 1 according to an example embodiment of the present disclosure;

(7) FIG. 6 illustrates a partial modified sectional view through a control body including a first connector portion according to an example embodiment of the present disclosure;

(8) FIG. 7 illustrates a sectional view through an extension of the first connector portion of FIG. 6 according to an example embodiment of the present disclosure;

(9) FIG. 8 illustrates a partial sectional view through a cartridge including a second connector portion according to an example embodiment of the present disclosure;

(10) FIG. 9 illustrates a partial modified sectional view through an aerosol delivery device including the control body of FIG. 6 and the cartridge of FIG. 8 according to an example embodiment of the present disclosure;

(11) FIG. 10 illustrates a perspective view of a receptacle including a center pin according to an example embodiment of the present disclosure;

(12) FIG. 11 illustrates a perspective view of an extension configured to engage the receptacle of FIG. 10 according to an example embodiment of the present disclosure;

(13) FIG. 12 illustrates a perspective view of an aerosol delivery device including a plurality of spring pins and a corresponding receptacle according to an example embodiment of the present disclosure; and

(14) FIG. 13 schematically illustrates a method for assembling an aerosol delivery device according to an example embodiment of the present disclosure.

#### DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(15) The present disclosure will now be described more fully hereinafter with reference to exemplary embodiments thereof. These exemplary embodiments are described so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Indeed, the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. As used in the specification, and in the appended claims, the singular forms “a”, “an”, “the”, include plural variations unless the context clearly dictates otherwise.

(16) The present disclosure provides descriptions of systems for assembling aerosol delivery devices. The aerosol delivery devices may use electrical energy to heat a material (preferably without combusting the material to any significant degree) to form an inhalable substance; such articles most preferably being sufficiently compact to be considered “hand-held” devices. An aerosol delivery device may provide some or all of the sensations (e.g., inhalation and exhalation rituals, types of tastes or flavors, organoleptic effects, physical feel, use rituals, visual cues such as those provided by visible aerosol, and the like) of smoking a cigarette, cigar, or pipe, without any substantial degree of combustion of any component of that article or device. The aerosol delivery device may not produce smoke in the sense of the aerosol resulting from by-products of combustion or pyrolysis of tobacco, but rather, that the article or device most preferably yields vapors (including vapors within aerosols that can be considered to be visible aerosols that might be considered to be described as smoke-like) resulting from volatilization or vaporization of certain components of the article or device, although in other embodiments the aerosol may not be visible. In highly preferred embodiments, aerosol delivery devices may incorporate tobacco and/or components derived from tobacco. As such, the aerosol delivery device can be characterized as an electronic smoking article such as an electronic cigarette or “e-cigarette.”

(17) While the systems are generally described herein in terms of embodiments associated with aerosol delivery devices such as so-called “e-cigarettes,” it should be understood that the mechanisms, components, features, and methods may be embodied in many different forms and associated with a variety of articles. For example, the description provided herein may be employed in conjunction with embodiments of traditional smoking articles (e.g., cigarettes, cigars, pipes, etc.), heat-not-burn cigarettes, and related packaging for any of the products disclosed herein. Accordingly, it should be understood that the description of the mechanisms, components, features, and methods disclosed herein are discussed in terms of embodiments relating to aerosol delivery mechanisms by way of example only, and may be embodied and used in various other products and methods.

(18) Aerosol delivery devices of the present disclosure also can be characterized as being vapor-producing articles or medicament delivery articles. Thus, such articles or devices can be adapted so as to provide one or more substances (e.g., flavors and/or pharmaceutical active ingredients) in an inhalable form or state. For example, inhalable substances can be substantially in the form of a vapor (i.e., a substance that is in the gas phase at a temperature lower than its critical point). Alternatively, inhalable substances can be in the form of an aerosol (i.e., a suspension of fine solid particles or liquid droplets in a gas). For purposes of simplicity, the term “aerosol” as used herein is meant to include vapors, gases and aerosols of a form or type suitable for human inhalation, whether or not visible, and whether or not of a form that might be considered to be smoke-like.

(19) In use, aerosol delivery devices of the present disclosure may be subjected to many of the

physical actions employed by an individual in using a traditional type of smoking article (e.g., a cigarette, cigar or pipe that is employed by lighting and inhaling tobacco). For example, the user of an aerosol delivery device of the present disclosure can hold that article much like a traditional type of smoking article, draw on one end of that article for inhalation of aerosol produced by that article, take puffs at selected intervals of time, etc.

(20) Smoking articles of the present disclosure generally include a number of components provided within an outer shell or body. The overall design of the outer shell or body can vary, and the format or configuration of the outer body that can define the overall size and shape of the smoking article can vary. Typically, an elongated body resembling the shape of a cigarette or cigar can be formed from a single, unitary shell; or the elongated body can be formed of two or more separable pieces. For example, a smoking article can comprise an elongated shell or body that can be substantially tubular in shape and, as such, resemble the shape of a conventional cigarette or cigar. However, various other shapes and configurations may be employed in other embodiments (e.g., rectangular or fob-shaped).

(21) In one embodiment, all of the components of the smoking article are contained within one outer body or shell. Alternatively, a smoking article can comprise two or more shells that are joined and are separable. For example, a smoking article can possess at one end a control body comprising a shell containing one or more reusable components (e.g., a rechargeable battery and various electronics for controlling the operation of that article), and at the other end and removably attached thereto a shell containing a disposable portion (e.g., a disposable flavor-containing cartridge). More specific formats, configurations and arrangements of components within the single shell type of unit or within a multi-piece separable shell type of unit will be evident in light of the further disclosure provided herein. Additionally, various smoking article designs and component arrangements can be appreciated upon consideration of the commercially available electronic smoking articles.

(22) Aerosol delivery devices of the present disclosure most preferably comprise some combination of a power source (i.e., an electrical power source), at least one controller (e.g., means for actuating, controlling, regulating and/or ceasing power for heat generation, such as by controlling electrical current flow from the power source to other components of the aerosol delivery device), a heater or heat generation component (e.g., an electrical resistance heating element or component commonly referred to as part of an “atomizer”), and an aerosol precursor composition (e.g., commonly a liquid capable of yielding an aerosol upon application of sufficient heat, such as ingredients commonly referred to as “smoke juice,” “e-liquid” and “e-juice”), and a mouthend region or tip for allowing draw upon the aerosol delivery device for aerosol inhalation (e.g., a defined airflow path through the article such that aerosol generated can be withdrawn therefrom upon draw).

(23) Alignment of the components within the aerosol delivery device of the present disclosure can vary. In specific embodiments, the aerosol precursor composition can be located near an end of the aerosol delivery device which may be configured to be positioned proximal to the mouth of a user so as to maximize aerosol delivery to the user. Other configurations, however, are not excluded. Generally, the heating element can be positioned sufficiently near the aerosol precursor composition so that heat from the heating element can volatilize the aerosol precursor (as well as one or more flavorants, medicaments, or the like that may likewise be provided for delivery to a user) and form an aerosol for delivery to the user. When the heating element heats the aerosol precursor composition, an aerosol is formed, released, or generated in a physical form suitable for inhalation by a consumer. It should be noted that the foregoing terms are meant to be interchangeable such that reference to release, releasing, releases, or released includes form or generate, forming or generating, forms or generates, and formed or generated. Specifically, an inhalable substance is released in the form of a vapor or aerosol or mixture thereof, wherein such terms are also interchangeably used herein except where otherwise specified.

(24) As noted above, the aerosol delivery device may incorporate a battery or other electrical power source (e.g., a capacitor) to provide current flow sufficient to provide various functionalities to the aerosol delivery device, such as powering of a heater, powering of control systems, powering of indicators, and the like. The power source can take on various embodiments. Preferably, the power source is able to deliver sufficient power to rapidly heat the heating element to provide for aerosol formation and power the aerosol delivery device through use for a desired duration of time. The power source preferably is sized to fit conveniently within the aerosol delivery device so that the aerosol delivery device can be easily handled. Additionally, a preferred power source is of a sufficiently light weight to not detract from a desirable smoking experience.

(25) More specific formats, configurations and arrangements of components within the aerosol delivery device of the present disclosure will be evident in light of the further disclosure provided hereinafter. Additionally, the selection of various aerosol delivery device components can be appreciated upon consideration of the commercially available electronic aerosol delivery devices. Further, the arrangement of the components within the aerosol delivery device can also be appreciated upon consideration of the commercially available electronic aerosol delivery devices.

(26) One example embodiment of an aerosol delivery device **100** is illustrated in FIG. **1**. In particular, FIG. **1** illustrates an aerosol delivery device **100** including a control body **200** and a cartridge **300**. The control body **200** and the cartridge **300** can be permanently or detachably aligned in a functioning relationship. Various connectors may connect the cartridge **300** to the control body **200** to result in a threaded engagement, a press-fit engagement, an interference fit, a magnetic engagement, or the like. The aerosol delivery device **100** may be substantially rod-like, substantially tubular shaped, or substantially cylindrically shaped in some embodiments when the cartridge **300** and the control body **200** are in an assembled configuration. However, various other configurations such as rectangular or fob-shaped may be employed in other embodiments.

(27) In specific embodiments, one or both of the cartridge **300** and the control body **200** may be referred to as being disposable or as being reusable. For example, the control body **200** may have a replaceable battery or a rechargeable battery and/or a capacitor and thus may be combined with any type of recharging technology, including connection to a typical alternating current electrical outlet, connection to a car charger (i.e., cigarette lighter receptacle), and connection to a computer, such as through a universal serial bus (USB) cable. Further, in some embodiments the cartridge **300** may comprise a single-use cartridge, as disclosed in U.S. Pat. No. 8,910,639 to Change et al., which is incorporated herein by reference in its entirety.

(28) FIG. **2** illustrates an exploded view of the control body **200** of the aerosol delivery device **100** according to an example embodiment of the present disclosure. As illustrated, the control body **200** may comprise a coupler **202**, an outer body **204** (i.e., a control body outer body), a sealing member **206**, an adhesive member **208** (e.g., KAPTON® tape), a flow sensor **210** (e.g., a puff sensor or pressure switch), a controller **212**, a spacer **214**, an electrical power source **216** (e.g., a battery, which may be rechargeable), a circuit board with an indicator **218** (e.g., a light emitting diode (LED)), a connector circuit **220**, and an end cap **222**. Examples of electrical power sources are described in U.S. Pat. App. Pub. No. 2010/0028766 by Peckerar et al., the disclosure of which is incorporated herein by reference in its entirety.

(29) With respect to the flow sensor **210**, representative current regulating components and other current controlling components including various microcontrollers, sensors, and switches for aerosol delivery devices are described in U.S. Pat. No. 4,735,217 to Gerth et al., U.S. Pat. Nos. 4,922,901, 4,947,874, and 4,947,875, all to Brooks et al., U.S. Pat. No. 5,372,148 to McCafferty et al., U.S. Pat. No. 6,040,560 to Fleischhauer et al., U.S. Pat. No. 7,040,314 to Nguyen et al., and U.S. Pat. No. 8,205,622 to Pan, all of which are incorporated herein by reference in their entireties. Reference also is made to the control schemes described in U.S. App. Pub. No. 2014/0270727 to Ampolini et al., which is incorporated herein by reference in its entirety.

(30) In one embodiment the indicator **218** may comprise one or more light emitting diodes. The

indicator **218** can be in communication with the controller **212** through the connector circuit **220** and be illuminated, for example, during a user drawing on a cartridge coupled to the coupler **202**, as detected by the flow sensor **210**. The end cap **222** may be adapted to make visible the illumination provided thereunder by the indicator **218**. Accordingly, the indicator **218** may be illuminated during use of the aerosol delivery device **100** to simulate the lit end of a smoking article. However, in other embodiments the indicator **218** can be provided in varying numbers and can take on different shapes and can even be an opening in the outer body (such as for release of sound when such indicators are present).

(31) Still further components can be utilized in the aerosol delivery device of the present disclosure. For example, U.S. Pat. No. 5,154,192 to Sprinkel et al. discloses indicators for smoking articles; U.S. Pat. No. 5,261,424 to Sprinkel, Jr. discloses piezoelectric sensors that can be associated with the mouth-end of a device to detect user lip activity associated with taking a draw and then trigger heating of a heating device; U.S. Pat. No. 5,372,148 to McCafferty et al. discloses a puff sensor for controlling energy flow into a heating load array in response to pressure drop through a mouthpiece; U.S. Pat. No. 5,967,148 to Harris et al. discloses receptacles in a smoking device that include an identifier that detects a non-uniformity in infrared transmissivity of an inserted component and a controller that executes a detection routine as the component is inserted into the receptacle; U.S. Pat. No. 6,040,560 to Fleischhauer et al. describes a defined executable power cycle with multiple differential phases; U.S. Pat. No. 5,934,289 to Watkins et al. discloses photonic-optronic components; U.S. Pat. No. 5,954,979 to Counts et al. discloses means for altering draw resistance through a smoking device; U.S. Pat. No. 6,803,545 to Blake et al. discloses specific battery configurations for use in smoking devices; U.S. Pat. No. 7,293,565 to Griffen et al. discloses various charging systems for use with smoking devices; U.S. Pat. No. 8,402,976 to Fernando et al. discloses computer interfacing means for smoking devices to facilitate charging and allow computer control of the device; U.S. Pat. No. 8,689,804 to Fernando et al. discloses identification systems for smoking devices; and WO 2010/003480 by Flick discloses a fluid flow sensing system indicative of a puff in an aerosol generating system; all of the foregoing disclosures being incorporated herein by reference in their entireties. Further examples of components related to electronic aerosol delivery articles and materials or components that may be used in the present article are disclosed in U.S. Pat. No. 4,735,217 to Gerth et al.; U.S. Pat. No. 5,249,586 to Morgan et al.; U.S. Pat. No. 5,666,977 to Higgins et al.; U.S. Pat. No. 6,053,176 to Adams et al.; U.S. Pat. No. 6,164,287 to White; U.S. Pat. No. 6,196,218 to Voges; U.S. Pat. No. 6,810,883 to Felter et al.; U.S. Pat. No. 6,854,461 to Nichols; U.S. Pat. No. 7,832,410 to Hon; U.S. Pat. No. 7,513,253 to Kobayashi; U.S. Pat. No. 7,896,006 to Hamano; U.S. Pat. No. 6,772,756 to Shayan; U.S. Pat. Nos. 8,156,944 and 8,375,957 to Hon; U.S. Pat. No. 8,794,231 to Thorens et al.; U.S. Pat. No. 8,851,083 to Oglesby et al.; U.S. Pat. Nos. 8,915,254 and 8,925,555 to Monsees et al.; and U.S. Pat. No. 9,220,302 to DePiano et al.; U.S. Pat. App. Pub. Nos. 2006/0196518 and 2009/0188490 to Hon; U.S. Pat. App. Pub. No. 2010/0024834 to Oglesby et al.; U.S. Pat. App. Pub. No. 2010/0307518 to Wang; WO 2010/091593 to Hon; and WO 2013/089551 to Foo, each of which is incorporated herein by reference in its entirety. A variety of the materials disclosed by the foregoing documents may be incorporated into the present devices in various embodiments, and all of the foregoing disclosures are incorporated herein by reference in their entireties.

(32) FIG. 3 illustrates the cartridge **300** in an exploded configuration. As illustrated, the cartridge **300** may comprise a base **302**, a control component terminal **304**, an electronic control component **306**, a flow director **308**, an atomizer **310**, a reservoir substrate **312**, an outer body **314** (i.e., a cartridge outer body), a mouthpiece **316**, a label **318**, and first and second heating terminals **320a**, **320b** according to an example embodiment of the present disclosure.

(33) In some embodiments the first and second heating terminals **320a**, **320b** may be embedded in, or otherwise coupled to, the flow director **308**. For example, the first and second heating terminals **320a**, **320b** may be insert molded in the flow director **308**. Accordingly, the flow director **308** and

the first and second heating terminals may be collectively referred to as a flow director assembly **322**. Additional description with respect to the first and second heating terminals **320a**, **320b** and the flow director **308** is provided in U.S. Pat. Pub. No. 2015/0335071 to Brinkley et al., which is incorporated herein by reference in its entirety.

(34) The atomizer **310** may comprise a liquid transport element **324** and a heating element **326**. The cartridge may additionally include a base shipping plug engaged with the base and/or a mouthpiece shipping plug engaged with the mouthpiece in order to protect the base and the mouthpiece and prevent entry of contaminants therein prior to use as disclosed, for example, in U.S. Pat. No. 9,220,302 to DePiano et al., which is incorporated herein by reference in its entirety.

(35) The base **302** may be coupled to a first end of the outer body **314** and the mouthpiece **316** may be coupled to an opposing second end of the outer body to substantially or fully enclose other components of the cartridge **300** therein. For example, the electronic control component **306**, the flow director **308**, the atomizer **310**, and the reservoir substrate **312** may be substantially or entirely retained within the outer body **314**. The label **318** may at least partially surround the outer body **314**, and optionally the base **302**, and include information such as a product identifier thereon. The base **302** may be configured to engage the coupler **202** of the control body **200** (see, e.g., FIG. 2). In some embodiments the base **302** may comprise anti-rotation features that substantially prevent relative rotation between the cartridge and the control body as disclosed in U.S. Pat. App. Pub. No. 2014/0261495 to Novak et al., which is incorporated herein by reference in its entirety.

(36) A reservoir may be configured to retain the aerosol precursor composition. For example, as described above, the reservoir may comprise the reservoir substrate **312**. However, the reservoir may comprise any other embodiment of a container or a material configured to hold an aerosol precursor composition.

(37) Representative types of aerosol precursor components and formulations are also set forth and characterized in U.S. Pat. No. 7,726,320 to Robinson et al.; U.S. Pat. No. 8,881,737 to Collett et al.; and U.S. Pat. No. 9,254,002 to Chong et al., and U.S. Pat. Pub. Nos. 2013/0008457 to Zheng et al.; 2015/0020823 to Lipowicz et al.; and 2015/0020830 to Koller, as well as WO 2014/182736 to Bowen et al, the disclosures of which are incorporated herein by reference. Other aerosol precursors that may be employed include the aerosol precursors that have been incorporated in the VUSE® product by R. J. Reynolds Vapor Company, the BLU product by Lorillard Technologies, the MISTIC MENTHOL product by Mistic Ecigs, and the VYPE product by CN Creative Ltd. Also desirable are the so-called “smoke juices” for electronic cigarettes that have been available from Johnson Creek Enterprises LLC. Embodiments of effervescent materials can be used with the aerosol precursor, and are described, by way of example, in U.S. Pat. App. Pub. No. 2012/0055494 to Hunt et al., which is incorporated herein by reference. Further, the use of effervescent materials is described, for example, in U.S. Pat. No. 4,639,368 to Niazi et al.; U.S. Pat. No. 5,178,878 to Wehling et al.; U.S. Pat. No. 5,223,264 to Wehling et al.; U.S. Pat. No. 6,974,590 to Pather et al.; U.S. Pat. No. 7,381,667 to Bergquist et al.; U.S. Pat. No. 8,424,541 to Crawford et al; and U.S. Pat. No. 8,627,828 to Strickland et al.; as well as US Pat. Pub. Nos. 2010/0018539 to Brinkley et al. and 2010/0170522 to Sun et al.; and PCT WO 97/06786 to Johnson et al., all of which are incorporated by reference herein.

(38) The reservoir substrate **312** may comprise a plurality of layers of nonwoven fibers formed into the shape of a tube encircling the interior of the outer body **314** of the cartridge **300**. Thus, liquid components, for example, can be sorptively retained by the reservoir substrate **312**. The reservoir substrate **312** is in fluid connection with the liquid transport element **324**. Thus, the liquid transport element **324** may be configured to transport liquid from the reservoir substrate **312** to the heating element **326** via capillary action or other liquid transport mechanism.

(39) As illustrated, the liquid transport element **324** may be in direct contact with the heating element **326**. As further illustrated in FIG. 3, the heating element **326** may comprise a wire defining a plurality of coils wound about the liquid transport element **324**. In some embodiments the heating

element **326** may be formed by winding the wire about the liquid transport element **324** as described in U.S. Pat. No. 9,210,738 to Ward et al., which is incorporated herein by reference in its entirety. Further, in some embodiments the wire may define a variable coil spacing, as described in U.S. Pat. App. Pub. No. 2014/0270730 to DePiano et al., which is incorporated herein by reference in its entirety. Various embodiments of materials configured to produce heat when electrical current is applied therethrough may be employed to form the heating element **326**. Example materials from which the wire coil may be formed include Kanthal (FeCrAl), Nichrome, Molybdenum disilicide (MoSi<sub>2</sub>), molybdenum silicide (MoSi), Molybdenum disilicide doped with Aluminum (Mo(Si,Al)<sub>2</sub>), graphite and graphite-based materials, and ceramic (e.g., a positive or negative temperature coefficient ceramic).

(40) However, various other embodiments of methods may be employed to form the heating element **326**, and various other embodiments of heating elements may be employed in the atomizer **310**. For example, a stamped heating element may be employed in the atomizer, as described in U.S. Pat. App. Pub. No. 2014/0270729 to DePiano et al., which is incorporated herein by reference in its entirety. Further to the above, additional representative heating elements and materials for use therein are described in U.S. Pat. No. 5,060,671 to Counts et al.; U.S. Pat. No. 5,093,894 to Deevi et al.; U.S. Pat. No. 5,224,498 to Deevi et al.; U.S. Pat. No. 5,228,460 to Sprinkel Jr., et al.; U.S. Pat. No. 5,322,075 to Deevi et al.; U.S. Pat. No. 5,353,813 to Deevi et al.; U.S. Pat. No. 5,468,936 to Deevi et al.; U.S. Pat. No. 5,498,850 to Das; U.S. Pat. No. 5,659,656 to Das; U.S. Pat. No. 5,498,855 to Deevi et al.; U.S. Pat. No. 5,530,225 to Hajaligol; U.S. Pat. No. 5,665,262 to Hajaligol; U.S. Pat. No. 5,573,692 to Das et al.; and U.S. Pat. No. 5,591,368 to Fleischhauer et al., the disclosures of which are incorporated herein by reference in their entireties. Further, chemical heating may be employed in other embodiments. Various additional examples of heaters and materials employed to form heaters are described in U.S. Pat. No. 8,881,737 to Collett et al., which is incorporated herein by reference, as noted above.

(41) A variety of heater components may be used in the present aerosol delivery device. In various embodiments, one or more microheaters or like solid state heaters may be used. Microheaters and atomizers incorporating microheaters suitable for use in the presently disclosed devices are described in U.S. Pat. No. 8,881,737 to Collett et al., which is incorporated herein by reference in its entirety.

(42) The first heating terminal **320a** and the second heating terminal **320b** (e.g., negative and positive heating terminals) are configured to engage opposing ends of the heating element **326** and to form an electrical connection with the control body **200** (see, e.g., FIG. 2) when the cartridge **300** is connected thereto. Further, when the control body **200** is coupled to the cartridge **300**, the electronic control component **306** may form an electrical connection with the control body through the control component terminal **304**. The control body **200** may thus employ the controller **212** (see, FIG. 2) to determine whether the cartridge **300** is genuine and/or perform other functions in conjunction with the electronic control component **306**. Further, various examples of electronic control components and functions performed thereby are described in U.S. Pat. App. Pub. No. 2014/0096781 to Sears et al., which is incorporated herein by reference in its entirety.

(43) Accordingly, the heating terminals **320a**, **320b** and the control component terminal **304** may be employed to form connections with the control body **200** (see, e.g., FIG. 2). For example, FIG. 4 illustrates an enlarged end view of the cartridge **300** at the base **302**. As illustrated, the first heating terminal **320a**, the second heating terminal **320b**, and the control component terminal **304** may extend to exposed positions within the base **302**. Thereby, the heating terminals **320a**, **320b** and the control component terminal **304** may be positioned for engagement with the control body **200**.

(44) By way of example, FIG. 5 illustrates an end view of the control body **200** at the coupler **202**. As illustrated, the control body **200** may include a plurality of electrical contacts **224a-c** respectively configured to contact the end of the control component terminal **304** and the ends of the heater terminals **320a**, **320b** (see, FIG. 3). The electrical contacts **224a-c** may be positioned at

differing radial distances from a central opening **226** through the coupler **202** and positioned at differing depths within the coupler. The depth and radius of each of the electrical contacts **224a-c** is configured such that the end of the control component terminal **304** and the ends of the heater terminals **320a**, **320b** respectively come into contact therewith when the base **302** (see, FIG. 3) and the coupler **202** are joined together to establish an electrical connection therebetween.

(45) In the illustrated embodiment the electrical contacts **224a-c** comprise circular metal bands of varying radii positioned at differing depths within the coupler **202** as described above. Each of the bands defines a major contact surface facing radially inwardly toward the central axis of the coupler **202**. The bands defining the electrical contacts **224a-c** are separated from one another by stepped surfaces of the body of the coupler **202**, which may be oriented perpendicularly to the radially facing major surfaces of the electrical contacts.

(46) As illustrated in FIG. 5, the coupler **202** may further comprise an anti-rotation mechanism **228** configured to prevent rotation of the control body **200** relative to the cartridge **300** (see, e.g., FIG. 3) when engaged therewith. The anti-rotation mechanism **228** may comprise a plurality of protrusions **230** and a plurality of recesses **232** alternately disposed about an outer periphery of the coupler **202**. As further illustrated in FIG. 5, a width of each of the protrusions **230** may increase from the connector end of the coupler **202** toward the outer body **204**. Conversely, a width of each of the recesses **232** may decrease from the connector end of the coupler **202** toward the outer body **204**.

(47) As illustrated in FIG. 4, the base **302** of the cartridge **300** may include protrusions **328** and recesses **330**. Thereby, when the base **302** is received inside the coupler **202** (see, e.g., FIG. 5), the protrusions **328** and the recesses **330** of the base may respectively engage the recesses **232** and the protrusions **230** (see, FIG. 5) of the coupler. Accordingly, when the base **302** of the cartridge **300** engages the coupler **202** of the control body **200** (see, e.g., FIG. 2), relative rotation between the control body and the base may be resisted.

(48) As illustrated in FIG. 4, the base **302** of the cartridge **300** may further include a groove **332**. Additionally, as illustrated in FIG. 5, the coupler **202** of the control body **200** may include one or more circumferential protrusions **234**. The circumferential protrusions **234** may engage the groove **332** to thereby provide releasable interlocking between the cartridge **300** and the control body **200** which may resist decoupling of the cartridge from the control body. Various other details with respect to the components configured for coupling the cartridge and the control body, are provided, for example, in U.S. Pat. App. Pub. No. 2014/0261495 to DePiano et al., which is incorporated herein by reference in its entirety.

(49) During use, a user may draw on the mouthpiece **316** of the cartridge **300** of the aerosol delivery device **100** (see, FIG. 1). This may pull air through an air inlet in the control body **200** (see, e.g., FIG. 2) or in the cartridge **300**. For example, as illustrated in FIG. 5, in one embodiment an air inlet **236** may be defined between the coupler **202** and the outer body **204** of the control body **200** (see, e.g., FIG. 2), as described in U.S. Pat. No. 9,220,302 to DePiano et al., which is incorporated herein by reference in its entirety. However, the flow of air may be received through other parts of the aerosol delivery device **100** in other embodiments. As noted above with respect to FIG. 3, in some embodiments the cartridge **300** may include the flow director **308**. The flow director **308** may be configured to direct the flow of air received from the control body **200** to the heating element **326** of the atomizer **310**.

(50) A sensor in the aerosol delivery device **100** (e.g., the flow sensor **210** in the control body **200**; see FIG. 2) may sense the puff. When the puff is sensed, the control body **200** may direct current to the heating element **326** through a circuit including the first heating terminal **320a** and the second heating terminal **320b**. Accordingly, the heating element **326** may vaporize the aerosol precursor composition directed to an aerosolization zone from the reservoir substrate **312** by the liquid transport element **324**. Thus, the mouthpiece **316** may allow passage of air and entrained vapor (i.e., the components of the aerosol precursor composition in an inhalable form) from the cartridge



**300** to a consumer drawing thereon.

(51) Various other details with respect to the components that may be included in the cartridge **300** are provided, for example, in U.S. Pat. Pub. No. 2015/0335071 to Brinkley et al., which is incorporated herein by reference in its entirety. Various components of an aerosol delivery device according to the present disclosure can be chosen from components described in the art and commercially available. Reference is made for example to the reservoir and heater system for controllable delivery of multiple aerosolizable materials in an electronic smoking article disclosed in U.S. Pat. App. Pub. No. 2014/0000638 to Sebastian et al., which is incorporated herein by reference in its entirety.

(52) In another embodiment substantially the entirety of the cartridge may be formed from one or more carbon materials, which may provide advantages in terms of biodegradability and absence of wires. In this regard, the heating element may comprise carbon foam, the reservoir substrate may comprise carbonized fabric, and graphite may be employed to form an electrical connection with the power source and the controller. An example embodiment of a carbon-based cartridge is provided in U.S. Pat. App. Pub. No. 2013/0255702 to Griffith et al., which is incorporated herein by reference in its entirety.

(53) As noted above, some embodiments of aerosol delivery devices include multiple pieces (e.g., a cartridge and a control component), which may releasably engage one another. As further described above, the cartridge may include terminals that engage electrical contacts comprising circular metal bands at the control body. However, connectors defining this configuration may suffer from certain detriments.

(54) For example, coupling of the control body to the cartridge may displace the terminals from their initial positions such that reliable connections may not be established. Further, the circular metal bands may shift out of position or wear over time such that a reliable connection may not be established. Additionally, the coupler of the control body may wear at the circumferential protrusions and/or the base of the cartridge may wear at the groove such that the security of the interlocking connection between the cartridge and the control body may be adversely affected. In this regard, the coupler and/or the base may comprise plastic materials, which may be prone to wear from repeated use. Further, when the groove and/or the circumferential protrusions wear such that the mechanical interlocking between the cartridge and the control body is adversely affected, the electrical connection between the terminals of the cartridge and the electrical contacts of the control body may also be adversely affected. Accordingly, wear on the mechanical connector portions of the aerosol delivery device may detrimentally affect both the mechanical and electrical connecting characteristics of the aerosol delivery device.

(55) Thus, the present disclosure is directed to alternative embodiments of aerosol delivery devices including connectors that may avoid some or all of the problems noted above. The aerosol delivery devices described hereinafter may include some or all of the components of the aerosol delivery devices described above. Accordingly, for brevity purposes each of the components of the aerosol delivery device described hereinafter may not be described or illustrated where the components described above, or components substantially corresponding to the components described above, may be employed.

(56) In this regard, FIG. **6** illustrates a partial, modified, sectional view through a control body **400** according to an embodiment of the present disclosure. As noted above, many of the components of the control body **400** may be the same as, or substantially the same as, the components of the control body **200**, and hence all of the components of the control body **400** are not shown or discussed in detail. In this regard, the control body **200** may include, amongst other components, a coupler **402**, an outer body **404** (i.e., a control body outer body), a flow sensor **406**, a controller **408**, and an electrical power source **410**.

(57) The flow sensor **406** may be configured to detect a pressure drop associated with a draw on a cartridge when such a cartridge is connected thereto. In this regard, the flow sensor **406** may be in

fluid communication with a pressure port **412** defined by the coupler **402**. The pressure port **412** may be in fluid communication with an air inlet. For example, in the illustrated embodiment a pressure seal **414** seals around the flow sensor **406** and connects to a pressure tube **416** that is in fluid communication with the pressure port at the coupler **402**. Thereby, when the control body **400** engages a cartridge, the flow sensor **406** may be in fluid communication with an air inlet, which may be defined in the cartridge, to detect a puff on the cartridge.

(58) The controller **408** may be electrically coupled to the electrical power source **410**. For example, a plurality of battery wires **418** or other electrical connectors may connect the controller **408** to the electrical power source **410**. Accordingly, the controller **408** may receive power from the electrical power source **410**, which can then be directed to a cartridge to produce an aerosol.

(59) In this regard, the control body **400** may include a first connector portion **420**, which may be configured to engage a cartridge as described below. The first connector portion **420** may include the coupler **402**. Additionally, the first connector portion **420** may include a seal such as an O-ring **422**. The O-ring **422** may be positioned at an outer surface of the coupler **402** in order to engage a cartridge as described below.

(60) Further, the first connector portion **420** may include an extension **424**. The extension **424** may be electrically coupled to the controller **408** via a plurality of extension wires **426a-c** or other electrical connectors. The extension **424** may be engaged with the coupler **402**. For example, the extension **424** may include a threaded section **428** which may screw into the coupler **402**. In this regard, the extension **424** may extend in an inner cavity **430** defined by the coupler **402**.

(61) As illustrated, in some embodiments an outer tip **432** of the extension **424** may terminate in the inner cavity **430** defined by the coupler **402**. Positioning the outer tip **432** of the extension **424** in the inner cavity **430** may protect the extension from damage and protect a user from contact with the extension. However, in other embodiments the extension may extend out of the inner cavity, or the coupler may not define an inner cavity and the extension may extend from the end of the coupler.

(62) As illustrated, the extension **424** may include a plurality of contact sections **434a-c** at the outer surface thereof. The contact section **434a-c** may be configured to form an electrical connection with a cartridge, as described below. The contact sections **434a-c** may be positioned (e.g., spaced apart) along a longitudinal length of the extension **424**. In this regard, the contact sections **434a-c** may be electrically insulated from one another. For example, the contact sections **434a-c** may be electrically insulated from one another by one or more spacers **436a**, **436b**. The spacers **436a**, **436b** may comprise an electrically insulating material such as plastic.

(63) Whereas FIG. 6 illustrates a side view of the extension **424**, FIG. 7 illustrates an enlarged sectional view through the extension **424** in order to show the components thereof. As illustrated, the third contact section **434c** may be defined by an inner contact **438**, which may extend centrally through the extension **424** along a longitudinal axis thereof. The second spacer **436b** may extend over a portion of the inner contact **438** such that the third contact section **434c** is exposed. A middle contact **440** may define the second contact portion **434b**. The inner contact **438** may extend through the middle contact **440** with the second spacer **436b** positioned therebetween to prevent the inner contact and the middle contact from contacting one another. An outer contact **442** may define the first contact section **434a**. The inner contact **438**, the middle contact **440**, and the spacers **436a**, **436b** may extend through the outer contact **442**. The first spacer **436a** may be positioned between the outer contact **442** and the middle contact **440** to prevent contact therebetween. Accordingly, each of the contacts **438**, **440**, **442** may be electrically insulated from each other.

(64) FIG. 8 illustrates a partial, modified, sectional view through a cartridge **500** according to an embodiment of the present disclosure. The cartridge **500** may include some or all of the components of the cartridge **300** (see, e.g., FIG. 3) described above. Many of the components of the cartridge **500** may be the same as, or substantially the same as, the components of the cartridge **300**, and hence all of the components of the cartridge **500** are not shown or discussed in detail. In

this regard, the cartridge **500** may include, amongst other components, a base **502**, a control component terminal **504**, an electronic control component **506**, a flow director **508**, a reservoir (e.g., a reservoir substrate **512**), and an outer body **514** (i.e., a cartridge outer body). The cartridge **500** may additionally include an atomizer, which may be substantially similar or identical to the atomizer **310** (see, e.g., FIG. 3), a mouthpiece, which may be substantially similar or identical to the mouthpiece **316** (see, e.g., FIG. 3), and a label, which may be substantially similar or identical to the label **318** (see, e.g., FIG. 3).

(65) As described below, the cartridge **500** may be configured to engage the control body **400** (see, e.g., FIG. 6) to form an electrical connection therewith. In this regard, the cartridge **500** may include a second connector portion **516**, which may be configured to engage the first connector portion **420** of the control body **400** (see, e.g., FIG. 6). The second connector portion **516** may include the base **502**. The second connector portion **516** may directly or indirectly engage the outer body **514**. For example, in the illustrated embodiment the flow tube **508** directly engages the outer body **514**, and the second connector portion **516** is engaged with the flow tube. In another embodiment the coupler and the flow tube may comprise an integral component that engages the outer body.

(66) The second connector portion **516** may include a receptacle **518**. The receptacle **518** may be configured to receive the extension **424** of the first connector portion **420** (see, FIG. 6) therein. Thereby, the receptacle **518** may establish an electrical connection with the extension **424**. In this regard, the receptacle **518** may include the coupler **502** and a plurality of terminals.

(67) In particular, the receptacle **518** of the second connector portion **516** may comprise a first heating terminal **520a** and a second heating terminal **520b**. The heating terminals **520a**, **520b** may be electrically coupled to the heating element of the atomizer. Further, as noted above, in some embodiments the second connector portion **516** may include the control component terminal **504**. The control component terminal **504** may be electrically coupled to the electronic control component **506**.

(68) FIG. 9 illustrates a partial, modified, sectional view through an aerosol delivery device **600** including the control body **400** of FIG. 6 and the cartridge **500** of FIG. 8. As illustrated, the first connector portion **420** and the second connector portion **516** may be configured to releasably engage each other. In this regard, the base **502** of the cartridge **500** may engage the coupler **402** of the control body **400**. For example, the coupler **402** of the control body **400** may be received in the base **502** of the cartridge **500**. In some embodiments one of the coupler and the base may include more circumferential protrusions (see, e.g., circumferential protrusions **234** in FIG. 5) and the other of the coupler and the base may include a groove (see, e.g., grooves **332** in FIG. 4) configured to engage the one or more circumferential protrusions. Thereby, mechanical interlocking may be provided by the protrusions and the groove. Further, in some embodiments the coupler and the base may include anti-rotation mechanisms as described above to thereby prevent relative rotation between the cartridge and the control body, which may reduce wear on the various connection mechanisms.

(69) In some embodiments the extension **424** may be configured to engage the receptacle **518** regardless of a relative rotational position of the cartridge **500** with respect to the control body **400**. For example, as illustrated, the extension **424** and the receptacle **518** may be centrally disposed with respect to a respective one of the first connector portion **420** and the second connector portion **516**. For example, the extension **424** and the receptacle **518** may be respectively aligned with central longitudinal axes of the control body **400** and the cartridge **500**. Thereby, a user may couple the cartridge **500** to the control body **400** without rotationally aligning the cartridge and the control body to expedite attachment thereof.

(70) Further, the cartridge **500** may electrically couple to the control body **400** when the first and second connector portions **420**, **516** engage one another. In this regard, as the base **502** of the cartridge **500** engages the coupler **402** of the control body **400**, the extension **424** may be received

in the receptacle **518**. When the extension **424** is received in the receptacle **518**, the heating terminals **520a**, **520b** and the control component terminal **504** may engage the extension. The first heating terminal **520a**, the second heating terminal **520b**, and the control component terminal **504** may contact differing sections of the extension **424**. In the illustrated embodiment the first heating terminal **520a** engages the first contact section **434a** of the extension **424**, the control component terminal **504** engages the second contact section **434b** of the extension, and the second heating terminal **520b** engages the third contact section **434c** of the extension. However, various other configurations may be employed in other embodiments.

(71) In some embodiments one or more of the terminals **520a**, **520b**, **504** may mechanically interlock with the extension **424**. In this regard, mechanical interlocking between one or more of the terminals **520a**, **520b**, **504** and the extension **424** may provide an improved connection between the cartridge **500** and the control body **400** and may provide a satisfying connected feel to a user. For example, the extension **424** may comprise an inwardly recessed detent **444** and the receptacle **518** may comprise a flexible member configured to engage the detent to retain the connection between the first connector portion **420** and the second connector portion **516**. In this regard, in the illustrated embodiment the second heating terminal **520b** comprises an end tab **522** configured to resiliently press into the detent **444**. In some embodiments mechanical interlocking between one or more of the terminals and the extension may be provided in combination with mechanical interlocking between the base and the coupler as described above, which may further provide a secure connection feel that may be desirable to a user. In other embodiments mechanical interlocking between one or more of the terminals and the extension may be provided as an alternative to interlocking between the base and the coupler.

(72) As noted above, each of the contact sections **434a-c** at the extension **424** may be electrically insulated from one another by the spacers **436a**, **436b**. Thereby, separate electrical connections may be formed between the first heating terminal **520a** and the first contact section **434a**, between the control component terminal **504** and the second contact section **434b**, and between the second heating terminal **520b** and the third contact section **434c**. Accordingly, the controller **408** may communicate with the electronic control component **506** to determine whether the cartridge **500** is genuine and/or perform other functions. In this regard, the second contact section **434b** may comprise a data contact section configured to form a data connection between the cartridge **500** and the control body **400**.

(73) Further, the controller **408** may direct current to the heating element of the atomizer through the heating terminals **520a**, **520b** when appropriate. In this regard, the flow sensor **406** may be configured to detect a puff on the cartridge **500**. When a user draws on the cartridge **500**, air may be directed into the aerosol delivery device **600** through an air inlet. As illustrated, in one embodiment an air inlet **524** may be defined in the cartridge **500**. For example, as illustrated, the air inlet **524** may be defined in the flow tube **508**. However, in other embodiments the air inlet may be defined in the base **502**, the outer body **514**, or other portion of the cartridge **500**.

(74) As the air enters the cartridge **500**, the flow sensor **406** may detect a pressure drop. In this regard, the pressure seal **414** seals around the flow sensor **406** and connects to a pressure tube **416** that is in fluid communication with the pressure port **412**. Further, the pressure port **412** may be in fluid communication with the air inlet **524**. For example, as illustrated, the pressure port **412** may extend to a cavity **446** positioned between the coupler **402** and the base **502** at an interior thereof. The cavity **446** may be substantially sealed due to the O-ring **422** or other sealing member being provided at an outer surface of the first connector portion **420** (e.g., at the outer surface of the coupler **402**) and configured to engage an inner surface of the second connector portion **516** (e.g., the inner surface of the base **502**) to form a seal between the base and the coupler. Additionally, the base **502** of the cartridge **500** may include a corresponding pressure port **526** that connects the cavity **446** to the air inlet **524**. Thereby, the flow sensor **406** may detect a pressure drop associated with a draw on the cartridge **500** through the pressure tube **416**, the pressure port **412**, the cavity

**446**, and the corresponding pressure port **526**.

(75) Accordingly, the controller **408** may direct current through a circuit to the heating element of the atomizer in the cartridge **500**. In this regard, the circuit may include one of the extension wires **426a**, the outer contact **442** (see, FIG. 7) defining the first contact section **434a**, the first heating terminal **520a**, the heating element, the second heating terminal **520b**, the inner contact **438** (see, FIG. 7) defining the third contact section **434c**, and an additional extension wire **426c**. Thereby, the heating element may heat the aerosol precursor composition stored in the reservoir substrate **512** or other reservoir to produce an aerosol that may be combined with the air, and which is directed to the user.

(76) Note that configuration of the aerosol delivery device **600** described above may provide certain benefits. In this regard, the airflow to the user may be separated from the electrical connectors that join at the first and second connector portions **420**, **516**. In this regard, the air inlet **524** is defined in the cartridge **500** such that the air does not flow through the connection between the cartridge and the control body **400**. Accordingly, any debris resulting from engagement and disengagement of the terminals **520a**, **520b**, **504** with the extension **424** may remain out of the airflow path.

(77) Further, although the flow sensor **406** is in fluid communication with the air inlet **524**, such fluid communication occurs along a substantially sealed path such that there is substantially no flow of air between the cartridge **500** and the control body. Additionally, the path through which the flow sensor **406** is in fluid communication with the air inlet **524** is separated from the extension **424** and the terminals **520a**, **520b**, **504**. Thereby, any debris resulting from engagement and disengagement of the terminals **520a**, **520b**, **504** with the extension **424** may not be drawn to the air inlet **524**.

(78) Further, the airflow path described above may isolate the electronic control component **506** and the controller **408** from the airflow. Thereby, issues with respect to the electronic components being damaged by moisture (e.g., by back puffs) may be avoided. This configuration also separates the electrical power source **410** from the airflow path such that issues with respect to chemicals or components of the electrical power source entering the airflow may be avoided.

(79) Additionally, the extension **424** described above may comprise a tip-ring-sleeve plug. Tip-ring-sleeve plugs are commonly employed as audio jacks to transmit and/or receive audio. In this regard, tip-ring-sleeve plugs may be configured to endure a relatively large number of engagement and disengagement cycles.

(80) However, the particular embodiment of the electrical connectors that may be employed in conjunction with the above-described aerosol delivery device may vary. For example, the particular configuration of the extension and the receptacle may vary. In this regard, FIG. 10 illustrates an alternate embodiment of the receptacle **518'** that may be employed in the cartridge **500** of the aerosol delivery device **600**. As illustrated, the receptacle **518'** may include a first heating terminal **520a'**, a second heating terminal **520b'**, and a control component terminal **504'**. A gap or spacer **521'** may be positioned between the first and second heating terminals **520a'**, **520b'**. The spacer **521'** may comprise a material that is not electrically conductive such that the first and second heating terminals **520a'**, **520b'** are electrically insulated from one another. As illustrated, the first and second heating terminals **520a'**, **520b'** may comprise rings that extend about inner surfaces of the receptacle **518'**. However, the control component terminal **504'** may comprise a center pin terminal that extends through the center of one or both of the first and second heating terminals **520a'**, **520b'**.

(81) In this regard, FIG. 11 illustrates an extension **424'** that may be employed in the control body **400** in embodiments wherein the cartridge includes the receptacle **518'** of FIG. 10. As illustrated, the extension **424'** may include a first contact section **434a'**, a second contact section **434b'**, and a third contact section **434c'**, which may be positioned in an inner cavity. A spacer **436a'** may be positioned between the first contact section **434a'** and the second contact section **434b'** to provide

electrical insulation therebetween. Further, a spacer **436b'** may be positioned between the third contact section **434c'** and the first and/or second contact sections **434a'**, **434b'** to provide electrical insulation therebetween. Thereby, the first and second heating terminals **520a'**, **520b'** may respectively form electrical connections with the first and second contact sections **434a'**, **434b'** and the control component terminal **504'** may form an electrical connection with the third contact section **434c'**. Accordingly, in some embodiments the connector may be substantially similar to an electrical connector employed in some personal electronic devices and may include some or all of the benefits of the tip-ring-sleeve plug described above.

(82) Additional embodiments of connectors may be employed in other embodiments of the present disclosure. For example, FIG. 12 illustrates an embodiment of the aerosol delivery device **600''** wherein the control body **400''** comprises a first connector portion **420''** comprising a plurality of extensions **424a-c''** respectively comprising one of a plurality of spring pins **434a-c''**, which may also be referred to as pogo pins. In this regard, the spring pins **434a-c''** may be spring-loaded so as to allow for firm engagement with corresponding terminals.

(83) For example, as further illustrated in FIG. 12, the cartridge **500''** may include a second connector portion **516''** comprising first and second heating terminals **520a''**, **520b''** and a control component terminal **504''**, which may be provided as rings and a center circle that may be separated from each other. The spring pins **434a-c''** may be positioned to engage a respective one of the terminals **520a''**, **520b''**, **504''** so as to form electrical connections therewith. Usage of spring pins may provide some or all of the benefits of the tip-ring-sleeve plug described above. Additionally, spring pins may be configured to endure a higher number of engagement and disengagement cycles due to the spring pins **434a-c''** not sliding against the terminals **520a''**, **520b''**, **504''** during contact therebetween.

(84) Note that although the portions of the connectors have been described above as being attached to one of the cartridge and the control body, the configurations described above have been provided by way of example only. In this regard, the portions of the connectors may be reversed between the cartridge and the control body such that, for example, the cartridge includes an extension and the control body includes a receptacle. In this regard, it may be desirable to attach the cheaper and/or more durable portion of the connector to the control body, which may be reusable whereas the cartridge may be disposable in some embodiments.

(85) In an additional embodiment a method for assembling an aerosol delivery device is provided. As illustrated in FIG. 13, the method may include forming a control body by inserting an electrical power source into a control body outer body and engaging a first connector portion with the control body outer body at operation **702**. Further, the method may include forming a cartridge by inserting a reservoir and an atomizer into a cartridge outer body and engaging a second connector portion with the cartridge outer body, the reservoir being configured to contain an aerosol precursor composition and the atomizer being configured to heat the aerosol precursor composition received from the reservoir to produce an aerosol. The first connector portion and the second connector portion may be configured to releasably engage each other. One of the first connector portion and the second connector portion may comprise an extension and the other of the first connector portion and the second connector portion may comprise a receptacle configured to receive the extension. The extension may comprise a plurality of contact sections positioned along a longitudinal length thereof. The contact sections may be electrically insulated from one another by at least one spacer and may be configured to form an electrical connection with the receptacle.

(86) In some embodiments of the method engaging the first connector portion with the control body outer body at operation **702** may include engaging a coupler with a flow tube and engaging the flow tube with the control body outer body. Engaging the second connector portion with the cartridge outer body at operation **704** may include engaging a base with the cartridge outer body. Forming the control body at operation **702** may further comprise engaging an O-ring with the coupler. The O-ring may be configured to engage an inner surface of the second connector portion.

(87) In some embodiments forming the control body at operation 702 may further comprise inserting a flow sensor in the control body outer body. The coupler may define a pressure port configured to be in fluid communication with the cartridge when the first connector portion engages the second connector portion. The method may further include engaging a pressure tube with the flow sensor and with the coupler.

(88) The method may further include engaging the extension with one of the coupler and the base and engaging the receptacle with the other of the coupler and the base. Engaging the extension with one of the coupler and the base and engaging the receptacle with the other of the coupler and the base may include centrally disposing the extension and the receptacle with respect to a respective one of the coupler and the base. Engaging the extension with one of the coupler and the base may include engaging a tip-ring-sleeve plug with one of the coupler and the base. The method may further include inserting a controller into the control body outer body and inserting an electronic control component into the cartridge outer body. Additionally, the method may include electrically coupling a data contact section of the extension with one of the controller and the electronic control component.

(89) Many modifications and other embodiments of the disclosure will come to mind to one skilled in the art to which this disclosure pertains having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed herein and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

## Claims

1. An electronic smoking article comprising: a cartridge comprising an outer body with an interior and defining a longitudinal axis extending through a center of the outer body between a first end and a second end thereof; a control body comprising a cavity; a base arranged at an open bottom surface of the cartridge for coupling the outer body and the control body; an air inlet defined in the outer body and extending to the interior thereof, the air inlet enabling air flow directly into the cartridge; a pressure port extending through the base to the air inlet, the pressure port arranged to connect the cavity in the control body with the air inlet; a flow sensor arranged within the smoking article to detect a pressure drop associated with a draw on the cartridge communicated through the pressure port, wherein the flow sensor is arranged within the cavity of the control body; a sealing member configured to form a seal around the flow sensor; and a pressure tube with a first end in fluid communication with the flow sensor and an opposing second end in fluid communication with the cavity, wherein the draw on the cartridge causing the air flow through the inlet aperture is communicated to the cavity through the pressure port and the pressure drop is detected by the flow sensor through the pressure tube.

2. The electronic smoking article of claim 1, further comprising a flow tube extending in the interior of the outer body coaxially with the longitudinal axis, the air inlet being in communication with the flow tube to direct air toward the flow tube after the draw on the cartridge.

3. The electronic smoking article of claim 1, wherein the seal around the flow sensor by the sealing member is configured as an air tight seal around the flow sensor.

4. The electronic smoking article of claim 3, further comprising a controller connected to the flow sensor and arranged within the air tight seal formed by the sealing member.

5. The electronic smoking article of claim 4, further comprising an electrical power source electrically coupled to the controller and within an interior of the control body, wherein the controller receives power from the electrical power source and directs power to the cartridge.

6. The electronic smoking article of claim 5, further comprising an atomizer within the interior of

the outer body, wherein the controller directs current to the atomizer.

7. The electronic smoking article of claim 6, wherein the control body comprises a first connector portion arranged to engage the cartridge, the first connector portion comprising an extension electrically coupled to the controller.

8. The electronic smoking article of claim 7, further comprising one or more heating terminals mechanically interlocking with the extension and arranged to engage the atomizer to form an electrical connection with the control body.

9. The electronic smoking article of claim 8, wherein each of the one or more heating terminals comprises a flexible portion and the extension comprises an inwardly recessed detent, and wherein the flexible portion of each of the one or more heating terminals is configured to resiliently press into the detent to form the electrical connection with the control body.

10. The electronic smoking article of claim 7, wherein the extension comprises one or more contact sections, at least one of the contact sections being a data contact section to form a data connection between the cartridge and the control body.

11. The electronic smoking article of claim 6, further comprising an aerosol precursor composition, the atomizer being configured to vaporize the aerosol precursor composition.

12. The electronic smoking article of claim 1, wherein the air inlet extends through the outer body substantially perpendicular to the longitudinal axis.

13. The electronic smoking article of claim 1, wherein the pressure port extends parallel to and offset from the longitudinal axis of the outer body.

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